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Course: programming fundamentals

Assignment: Assignment LAB 1

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QUESTION 1 SUM OF ODD NUMBERS

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-d
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java
 */

/**
 *
 * @author PC
 */
import java.util.Scanner;
public class LaibaMudassarQ1 {
  public static void main(String[]args) {
      Scanner sc=new Scanner (System.in);
      System.out.println("Enter a number");
      int num=sc.nextInt();
      int sum=0;
      for(int i=1;i<=num;i++){
         if(i%2==0) continue;
                  if(i%10==7) continue;
              sum+=i;
              if (sum >500)break;
      }
      System.out.println("Final sum"+sum);
      }
  }
```

OUTPUT

```
run:
Enter a number
20
Final sum76
BUILD SUCCESSFUL (total time: 3 seconds)
```

QUESTION NO 2 TRAIN TICKET PRICE

```java
import java.util.Scanner;
public class LaibaMudassarQ2 {
   public static void main(String[]args){
    Scanner sc=new Scanner (System.in);
      System.out.println("enter age");
      int age=sc.nextInt();
      System.out.println("Enter distance in km ");
      int distance=sc.nextInt();
     int pricePerKm;
      switch(age/12){
          case 0:
           pricePerKm=20;
           break;
          case 1:
          case 2:
          case 3:
          case 4:
              pricePerKm=50;
              break;
          default:
              pricePerKm=10;
      }
      int totalPrice=pricePerKm*distance;
      System.out.println("total ticket price= "+totalPrice);
   }
}
```

OUTPUT

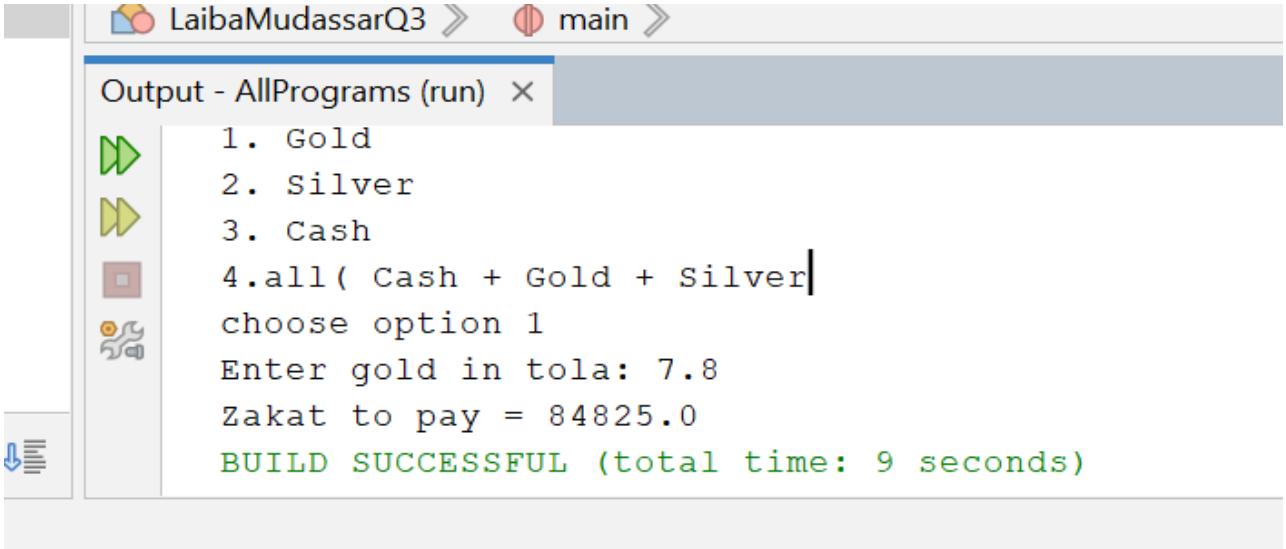
```
run:
enter age
24
Enter distance in km
55
total ticket price= 2750
BUILD SUCCESSFUL (total time: 9 seconds)
```

QUESTION NO 3 ZAKAT CALCULATOR

```java
import java.util.Scanner;
public class LaibaMudassarQ3 {
        public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.println("1. Gold");
        System.out.println("2. Silver");
        System.out.println("3. Cash");
        System.out.println("4.all( Cash + Gold + Silver");
        System.out.print("choose option ");
        int choice = input.nextInt();
        double zakat = 0;
        switch(choice) {
            case 1:
                System.out.print("Enter gold in tola: ");
                double gold = input.nextDouble();
                if(gold >= 7.5) {
                    double totalGold = gold * 435000;
                    zakat = totalGold * 0.025;
                }
                break;
            case 2:
                System.out.print(("Enter silver in tola: "));
                double silver = input.nextDouble();
                if(silver >= 52) {
                    double totalSilver = silver * 5500;
                    zakat = totalSilver * 0.025;
                }
```

```java
                break;
            case 3:
                System.out.print("Enter cash: ");
                double cash = input.nextDouble();
                if(cash >= 180000) {
                    zakat = cash * 0.025;
                }
                break;
            case 4:
                System.out.print("Enter cash: ");
                double c = input.nextDouble();
                System.out.print("Enter gold in tola: ");
                double g = input.nextDouble();
                System.out.print("Enter silver in tola: ");
                double s = input.nextDouble();
                double total = c + (g * 435000) + (s * 5500);
                if(total >= 180000) {
                    zakat = total * 0.025;
                }
                break;
            default:
                System.out.println("Invalid choice");
        }
        System.out.println("Zakat to pay = " + zakat);
    }
}
```

OUTPUT


LaibaMudassarQ3
main
Output - AllPrograms (run)
1. Gold
2. Silver
3. Cash
4.all(Cash + Gold + Silver
choose option 1
Enter gold in tola: 7.8
Zakat to pay = 84825.0
BUILD SUCCESSFUL (total time: 9 seconds)

QUESTION NO 4 NUMBER GUEESING

```java
/**
 *
 * @author PC
 */
import java.util.Scanner;
public class LaibaMudassarQ4 {
    public static void main(String[]args){
        Scanner input=new Scanner(System.in);
        int num=50;
        int attempts=5;
        System.out.println("guess a number between 1 and 100");
        while(attempts>0){
         System.out.println("enter your guess");
         int guess=input.nextInt();
         if(guess==num){
             System.out.println("Correct! you win");
         }
        }
    }
}
```

OUTPUT

```
Output - AllPrograms (run) ...
run:
guess a number between 1 and 100
enter your guess
34
enter your guess
56
enter your guess
57
enter your guess
48
enter your guess
50
Correct! you win
enter your guess
```

QUSTION NO 5 PANLINDROME NUMBER

```java
/**
 *
 * @author PC
 */
import java.util.Scanner;
public class LaibaMudassarQ5 {
    public static void main(String[]args){
      Scanner sc=new Scanner (System.in);
        System.out.println("Enter three digit number");
        int num=sc.nextInt();
        int a=num/100;
        int c=num%10;
        if(a==c)
        {  System.out.println("palidrome");
                }
        else
        {
            System.out.println(" not a palidrome");
        }
    }
}
```

OUTPUT

```
Output
AllPrograms (run)    AllPrograms (run) #2
run:
Enter three digit number
222
palidrome
BUILD SUCCESSFUL (total time: 7 seconds)
```

QUESTION NO 6 COUNT POSITIVE NEGATIVR AVERAGE

Source History

Source View

```java
import java.util.Scanner;
public class LaibaMudassarQ6 {
  public static void main(String[]args){
      Scanner sc=new Scanner (System.in);
      int positiveCount=0,negativeCount=0,sum=0;
      double totalNumbers=0;
      System.out.println("enter number (0 to stop)");
      while(true) {
            int num = sc.nextInt();
            if(num == 0) {
                break;  // stop input if user enters 0
            }
            if(num > 0) {
                positiveCount++;
            } else {
                negativeCount++;
            }
            sum += num;
            totalNumbers++;
        }
        double average = (double) sum/ totalNumbers;
        System.out.println("Number of positive numbers: " + positiveCount);
        System.out.println("Number of negative numbers: " + negativeCount);
        System.out.println("Average of numbers: " + average);
    }
}
```

LaibaMudassarQ6 > main > while (true) >

OUTPUT

AllPrograms (run) AllPrograms (run) #2

```
run:
enter number (0 to stop)
1
2
3
0
Number of positive numbers: 3
Number of negative numbers: 0
Average of numbers: 2.0
BUILD SUCCESSFUL (total time: 10 seconds)
```

QUESTION NO 7 HIGHEST SCORE

```java
*/
import java.util.Scanner;
public class LaibaMudassarQ7 {
 public static void main(String[]args){
     Scanner sc=new Scanner(System.in);
     System.out.println("Enter a numbr of students");
     int studentNumber=sc.nextInt();
     String topName="";
     int maxScore=0;
     for(int i=0;i<studentNumber;i++){
         System.out.println("enter name ");
         String name=sc.next();
          System.out.println("enter score ");
         int score=sc.nextInt();
         if(score > maxScore)
         {
           maxScore=score;
           topName=name;
         }
     }
      System.out.println("top student"+topName);

     }
    }
```

OUTPUT

```
run:
Enter a numbr of students
3
enter name
ali
enter score
67
enter name
bisma
enter score
57
enter name
amna
enter score
89
top studentamna
BUILD SUCCESSFUL (total time: 30 seconds)
```